Storm transposition

CEVE 543

2026-10-26

Summary

An event set is a long list of plausible storms with a probability attached to each. In this session we compute risk measures from an event set, build one for a watershed by moving observed storms around its region (stochastic storm transposition), weight storms that come with return periods, and see what a single rainfall number per storm leaves out.

- Average annual loss $\mathrm{AAL} = E[L]$, estimated by $\frac{1}{n}\sum_{i=1}^{n} L_i$ over n equally likely synthetic years, where L_i is the total loss in synthetic year i.
- VaR_α is the annual loss exceeded with probability $1-\alpha$; TVaR, also called CVaR or expected shortfall, is $E[L \mid L \geq \mathrm{VaR}_\alpha]$.
- From a ladder of design events (the 2- to 100-year storms, each with its damage d_k), AAL is the area under damage against annual exceedance probability $p_k = 1/T_k$, approximated with trapezoids: $\mathrm{AAL} \approx \sum_k (p_k - p_{k+1})\frac{d_k + d_{k+1}}{2}$.
- From a set of storms with no synthetic years, fit the distribution of an intensity measure, sort the storms from the rarest, $T_1 > T_2 > \cdots$, and weight each by its own rate $\lambda_j = 1/T_j - 1/T_{j-1}$ with $1/T_0 = 0$; then $\mathrm{AAL} = \sum_j \lambda_j d_j$.
- An intensity measure is one number per storm, and its frequency need not match the frequency of flooding or damage.

For supplementary notes, read:

- O'Rourke & Burdick (2025), pp. 9-10, on the exceedance probability curve, VaR, and TVaR.
- Wright et al. (2020) §2-3, pp. 2-5, on the transposition procedure and its assumptions.
- Baer et al. (2026), pp. 1-7, this week's reading.

Bibliography

Baer, J. A., Sebastian, A., Grimley, L. E., Doss-Gollin, J., Wright, D. B., Hussain, M. A., & Webber, M. K. (2026). Neglecting Spatiotemporal Rainfall Variability Misrepresents Flood Hazard and Risk. *Npj Natural Hazards*. https://doi.org/10.1038/s44304-026-00208-5

O'Rourke, C., & Burdick, J. (2025). *Catastrophe Modeling 101* (White Paper). Retrieved from https://www.nb.com/

Wright, D. B., Yu, G., & England, J. F. (2020). Six Decades of Rainfall and Flood Frequency Analysis Using Stochastic Storm Transposition: Review, Progress, and Prospects. *Journal of Hydrology*, *585*, 124816. https://doi.org/10.1016/j.jhydrol.2020.124816